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Enhanced Drought Prediction in Ethiopia: A Deep Learning Approach

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We aim to contribute to new global and national data and evidence that governments, decision-makers, citizens and researchers can use to improve people's lives and support the world's poorest people in their efforts to escape extreme poverty.

We are a consortium of the Universities of Cornell, Copenhagen, and Southampton led by Oxford Policy Management, in partnership with the World Bank's Development Data Group and funded by the UK Foreign, Commonwealth & Development Office.

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The DEEP Challenge Fund provides small to medium-sized grants to national researchers working on country-specific poverty reduction. The scope of the Fund was co-designed with Ethiopian policy and research communities at multiple stages.

Awards were selected through a fair and rigorous competitive process, led by a nationally constituted Technical Review Panel in collaboration with the Ethiopia National Steering Committee.

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Summary

Drought remains a multifaceted and devastating environmental hazard in the arid and semi-arid lowlands of Ethiopia, including in Afar and Somali regions, which experience extreme climatic vulnerability. This study investigates drought dynamics across six representative districts (woredas) by integrating remote sensing and climate-based metrics, including the Normalised Difference Vegetation Index (NDVI), Temperature Condition Index (TCI), Vegetation Condition Index (VCI), Vegetation Health Index (VHI), Standardised Precipitation Evapotranspiration Index (SPEI), and Composite Drought Index (CDI). Analysis of these indices reveals distinct spatio-temporal patterns characterised by persistent vegetation stress and a lack of ecological recovery. The observed variability, ranging from moderate resilience to acute drought, underscores the limitations of traditional statistical methods and classical machine learning in modelling complex, non-linear thermal-climatic interactions. To address these gaps, we conducted a comparative evaluation of three gradient-boosting frameworks (XGBoost, CatBoost and Light gradient-boosting machine (LightGBM)) against three deep learning architectures (Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM), Bidirectional Long Short-Term Memory (Bi-LSTM), and Transformer). The CNN-LSTM hybrid model demonstrated superior predictive performance, achieving a mean absolute error (MAE) of 0.28, a root mean square error (RMSE) of 0.38, and a coefficient of determination (R2) of 0.82. These results indicate that the integration of convolutional layers for spatial feature extraction with LSTM units for temporal dependencies significantly enhances the modelling of moisture-stressed lowland environments. This research provides a methodological framework for developing advanced, deep learning-based early warning systems to mitigate the socio-economic impacts of drought in highly vulnerable regions.

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1. Introduction

Ethiopia ranks among the top 10 nations that are most vulnerable to climate disasters (Gašparović, 2025). The country has experienced multiple climate-related shocks, including droughts, floods, landslides, pest infestations, and earthquakes, which have pushed many households below the poverty line. Over the past two decades, recurrent droughts and flash floods have had severe consequences for agricultural, agro-pastoral, and pastoral communities (Climate Change Laws of the World, 2022) The increasing frequency of these climate shocks worsens food shortages, with over 11 million people facing severe food insecurity in drought-affected areas (Gašparović, 2025). Vulnerable groups include persons with disabilities, older adults, people in remote areas, women and girls, and those with low literacy levels (IGAD, 2023). Droughts and floods are particularly impactful for these groups in Ethiopia (Cherinet *et al.*, 2022).

Drought is characterised by a prolonged deficit in precipitation that results in widespread water scarcity across multiple sectors (Wilhite, 2000; Yevjevich, 1967). In agrarian economies, these events create severe socio-economic challenges by reducing agricultural productivity, depleting water resources, and worsening food insecurity (Mishra and Singh, 2010; World Bank, 2016). Recent analyses indicate that poverty vulnerability in Ethiopia is significantly higher in the drought-prone lowlands than in any other ecological zone (Skoufias *et al*., 2021).

According to the World Bank's analysis of the 2016 Household Consumption Expenditure and Welfare Monitoring Surveys (World Bank, 2020), these lowlands cover more than 50% of the country's landmass but house only 12% of the population. In the arid and semi-arid areas, where livelihoods depend almost exclusively on rain-fed agriculture and pastoralism, recurrent droughts over the past two decades have had devastating effects on human well-being (Climate Change Laws of the World, 2022; Skoufias *et al.*, 2021).

Afar and Somali regions have been the most severely impacted; Somali Region recently endured five consecutive failed rainy seasons, displacing over a million people, while the collapse of pastoralist livelihoods in Afar has triggered critical levels of malnutrition (ACAPS, 2023). While accurate forecasting is essential for robust early warning and monitoring systems (Climate Change Laws of the World, 2022), drought remains inherently non-linear and complex. Consequently, traditional statistical methods, such as ARIMA, often fail to capture the intricate dependencies of such multi-scalar environmental crises (Nourani *et al.*, 2009).

To address these complexities, the research frontier has shifted towards sophisticated computational intelligence. Advancements in deep learning, and specifically LSTM networks for temporal memory (Hochreiter and Schmidhuber, 1997) and Transformers for attention-based weighting (Vaswani, 2017), have opened up new avenues for modelling high-dimensional environmental data. In Ethiopia, where climate variables are highly localised and influenced by varied topography, researchers have increasingly adopted these robust computational tools to improve disaster preparedness. Melese *et al*. (2025) utilised the Palmer Drought Severity Index (PDSI), alongside high-resolution TerraClimate data, to provide granular spatial insights into moisture deficits. Similarly, Turhan (2021) demonstrated that artificial neural network (ANN) architectures simulate complex rainfall–runoff relationships more effectively than traditional linear regression. Building on these advancements, integrated remote sensing indices – specifically VCI, TCI, and VHI – with ANN techniques to enhance agricultural drought prediction (Alemu and Zimale, 2025). Furthermore, Bojer *et al*. (2024) identified the CatBoost algorithm as a superior predictor for the Standardised Precipitation Index (SPI), highlighting the shift towards high-performance gradient-boosting in climatic modelling.

Despite this progress, the adoption of advanced modelling in the Ethiopian context remains fragmented. Most local studies continue to rely on isolated indices (e.g. SPI, NDVI, or PDSI), which fail to capture the synergistic interactions of drought, which is a phenomenon that is driven by interconnected meteorological, agricultural, and ecological processes (Mishra and Singh, 2010).

Consequently, even where machine learning is applied, high-order deep learning architectures remain underutilised and poorly optimised for the unique spatio-temporal complexities of the highly vulnerable Afar and Somali regions. This creates a critical gap in capturing the non-linear feedbacks between thermal stress and vegetation health.

This paper addresses these gaps by developing an integrated deep learning model utilising climate indices. The contributions of this work are as follows:

- The development of a robust framework that synthesises diverse environmental indices, including NDVI, VCI, TCI, VHI, and SPEI, to capture the multi-dimensional nature of drought.
- The optimisation of a deep learning model to address the complex, non-linear interactions inherent in drought forecasting.
- The establishment of a performance benchmark, by evaluating classical gradient-boosting frameworks (XGBoost, CatBoost, LightGBM) against advanced deep learning paradigms (CNN-LSTM, Bi-LSTM, and Transformer).

The remainder of this paper is organised as follows. Section 2 outlines the research design and methodology, detailing data pre-processing, feature engineering, and the machine learning framework. Section 3 presents the formulation and interpretation of the CDI. This is followed by the presentation of drought classification criteria in Section 4. Section 5 discusses model development, and Section 6 discusses performance evaluation criteria. Sections 7 and 8 present the exploratory analysis, and Section 9 details the final model performance results. Finally, Section 10 concludes the study by key findings and discussing their implications for future research.

2. Research design and methodology

This study focuses on six drought-prone administrative districts (woredas) located in the arid and semi-arid lowlands of eastern Ethiopia. These sites were purposively selected to capture a broad range of climate vulnerability and dominant livelihood systems across two major regions: Afar and Somali. Three woredas were chosen in Afar Region, an area characterised by extreme heat, sparse vegetation, and chronic water scarcity:

- Elidar: a border district where community survival depends heavily on traditional mobility and shared water wells;
- Ewa: a predominantly pastoral district that experiences increasing livestock losses due to highly erratic rainfall; and
- Bidu: in the north-western corner of the region, which faces compounded stress on pastoral livelihoods from frequent drought and localised volcanic activity.

The approximate land areas of Elidar, Ewa, and Bidu are 11,600 km², 1,463 km², and 4,800 km², respectively (Ethiopian Statistics Service, 2007).

Three woredas were selected in Somali Region, which is highly exposed to recurrent droughts and flash floods:

- Fik: situated in the northern Erer zone and characterised by remote pastoral communities that are dependent on seasonal rangelands;
- Gode: a more urbanised regional hub along the Shebelle River that remains vulnerable to seasonal drought and flash flooding; and
- Hargele: an isolated area with rugged terrain and minimal infrastructure, where communities primarily practise pastoralism under harsh climatic conditions.

The approximate land areas of Fik, Gode, and Hargele are 10,000 km², 8,200 km², and 6,400 km², respectively (Ethiopian Statistics Service, 2007). The geographical locations and distributions of these study sites are shown in Figure 1.

Figure 1: Administrative map of Ethiopia: Selected woredas in Afar and Somali regions

2.1. Data collection

To develop the composite drought prediction model, we integrated diverse datasets from climatic and remote sensing sources, as shown in Figure 2. Primary data sources included Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) (Funk *et al*., 2015), Moderate Resolution Imaging Spectro-radiometer (MODIS) satellite imagery (Earth Science Data Systems, 2025), and historical drought recurrence records from the Ethiopian Disaster Risk Management Commission (EDRMC). The temporal scope was limited to 2000–2022 to ensure the use of fully calibrated records and to maintain a consistent baseline. This approach avoids including preliminary or near-real-time datasets that have not undergone final quality control and validation by the relevant authorities, such as the MODIS Collection 6.1 reprocessing and official EDRMC reports. To validate the model's predictive accuracy, we used a ground truth dataset. The primary validation data were obtained from the EDRMC, specifically using the results of the multi-agency Belg (February–May) and Meher (June–September) seasonal assessments. These field-based reports provide a verified record of droughts and their impacts on pastoralist and agricultural livelihoods. We used these records to benchmark the model's pixel-level outputs against documented real-world drought events, ensuring that the predicted thresholds correspond to observed drought levels in the selected areas of Afar and Somali regions.

2.2. Data pre-processing and feature engineering

To achieve pixel-level fusion and temporal consistency across the six selected districts in Afar and Somali regions of Ethiopia, the data were spatially harmonised using a grid-based resampling approach to a uniform 5 km × 5 km resolution. This 5 km × 5 km grid, chosen to align all multi-source data ranging from 250 m to 5 km and to match the native resolution of the CHIRPS dataset (Funk *et al*., 2015), serves as the primary spatial unit for modelling. To mitigate the loss of spatial detail during the downscaling of the 250 m MODIS imagery (Earth Science Data Systems, 2025) an area-weighted averaging method was used. The resulting scale is highly relevant for regional drought monitoring in Ethiopia's arid regions. Although results are reported at the woreda (district) and regional levels for policy relevance, these represent the aggregate mean of all grid cells within the respective administrative boundaries. This approach ensures that pastoralist movements and agricultural impacts are assessed at the appropriate administrative level while maintaining modelling at the grid cell level. Missing satellite values were addressed using linear interpolation, and all features were aggregated to a monthly temporal scale before model training.

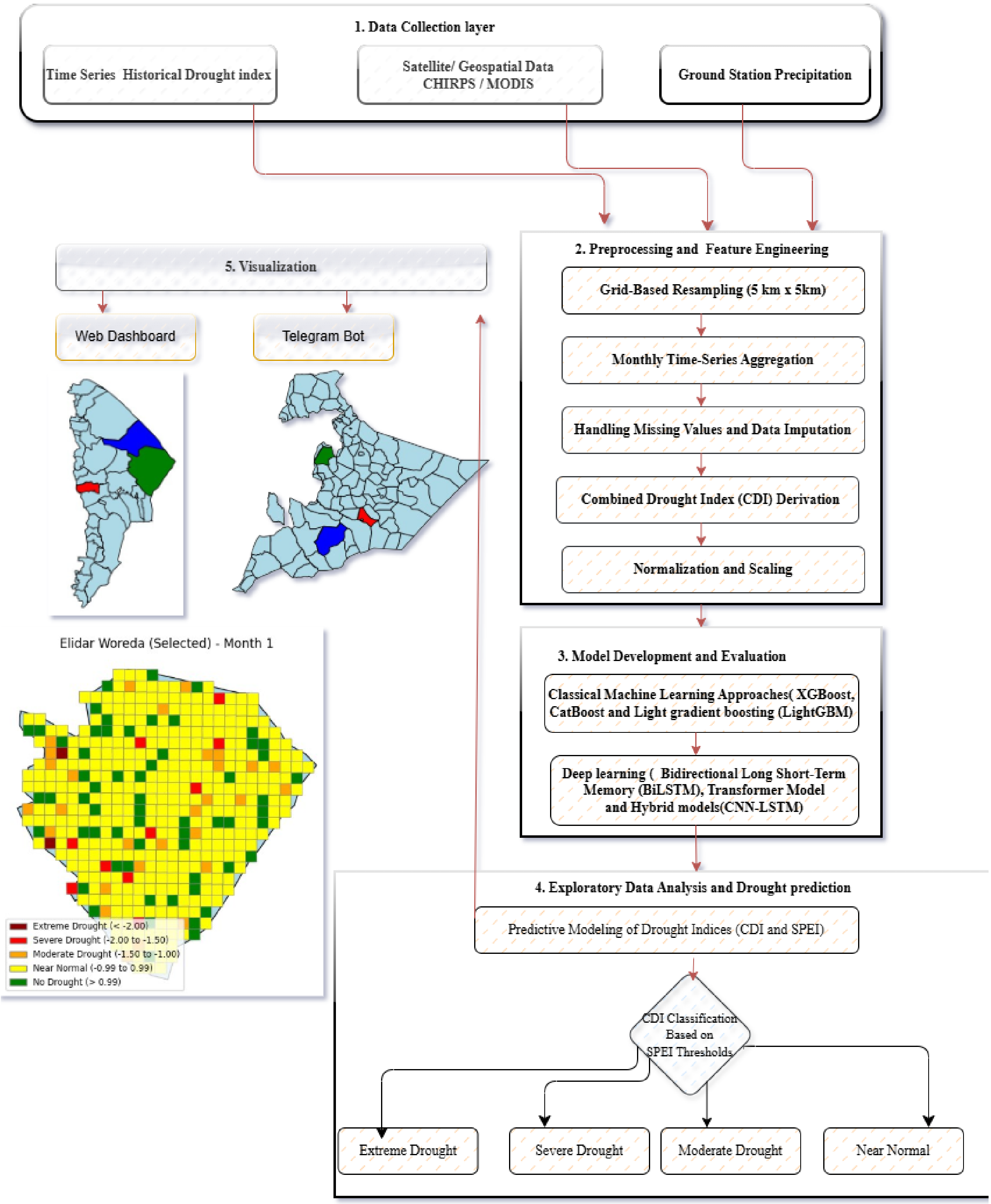
The collected datasets were pre-processed for subsequent analysis. Using climatic and remote sensing sources, we calculated vegetation (NDVI, TCI, VCI, and VHI) and meteorological (SPEI) drought indices.

- The NDVI is the standard remote sensing measure for quantifying the presence and condition of green vegetation (Rouse *et al*., 1974). It exploits the difference between the high reflectance of healthy vegetation in the near-infrared (NIR) band and its strong absorption in the red (RED) band. The NDVI is defined in Equation 1 .

$$NDVI = \frac{(NIR-RED)}{(NIR+RED)} \qquad \text{Equation 1}$$

  The NDVI was calculated using the MODIS/006/MOD13Q1 product (250 m spatial resolution, 16-day composite) (Earth Science Data Systems, 2025).

Figure 2: Methodological framework for multi-source geospatial data fusion and machine learning and deep learning



The workflow integrates multi-source data (CHIRPS, MODIS, and Ground Stations) resampled to a uniform 5 km × 5 km grid, which serves as the primary spatial unit for modelling. Predictive modelling and CDI derivation are performed at the pixel level, and results are aggregated to woreda and regional scales to support drought monitoring and visualisation on web and mobile platforms.

- TCI is calculated by normalising the land surface temperature (LST) at a specific time (τ) relative to its historical minimum (LST_{min}) and maximum (LST_{max}) for that period

(Kogan, 1995a). The TCI quantifies temperature stress, with lower values indicating higher heat stress. The TCI is defined in Equation 2:

$$\mathrm{TCI} = 100 \times \frac{\mathrm{LST_{min}} - \mathrm{LST_{\tau}}}{\mathrm{LST_{max}} - \mathrm{LST_{min}}} \qquad \text{Equation 2}$$

- VCI measures the current health of vegetation by comparing the observed NDVI (NDVI) to the historical NDVI range for that location and time. It functions as an anomaly index, where a score of 100 indicates optimal historical conditions and 0 indicates extreme stress. The calculation follows the method established by Kogan (1995a) (see Equation 3).

$$\mathrm{VCI} = 100 \times \frac{\mathrm{NDVI_{\tau}} - \mathrm{NDVI_{min}}}{\mathrm{NDVI_{max}} - \mathrm{NDVI_{min}}} \qquad \text{Equation 3}$$

- The VHI is a combined index that is used to observe drought conditions and vegetation stress by integrating the VCI and the TCI. The VHI is defined in Equation 4 as:

$$VHI = \alpha * VCI + (1 - \alpha) * TCI \qquad \text{Equation 4}$$

Where α (alpha) is weighting factor (commonly 0.5 for equal weighting).

- SPEI is a drought determination method based on precipitation and temperature (Vicente-Serrano *et al.*, 2010). SPEI is computed using precipitation data from the CHIRPS (v2.0) dataset. The index is derived by first calculating the monthly climatic water balance (D), which is the difference between monthly precipitation (P) and monthly potential evapotranspiration (PET) (see Equation 5).

$$D = P - PET \qquad \text{Equation 5}$$

The D series is then aggregated over the defined timescale (e.g. three months or six months) and fitted to a three-parameter log-logistic probability distribution to transform the aggregated values into standardised SPEI values.

We integrated remote sensing indices, specifically NDVI, TCI, VHI, and VCI (Kogan, 1995a), with climate data such as the SPEI. To ensure spatial and temporal alignment, all variables were aggregated to a monthly sampling frequency. For the SPEI, we utilised three-month cumulative values to represent the moisture deficit or surplus. This sliding window approach aligns the three-month climatic signal with discrete monthly vegetation observations (NDVI, VCI, and TCI), consistent with standard procedures for multi-scalar drought indices (Vicente-Serrano *et al*., 2010).

3. CDI formulation and interpretation

To capture the multifaceted nature of drought, this study integrates meteorological, agricultural, vegetation, and thermal indicators into CDI. This framework incorporates the SPEI to assess climatic water balance, alongside the NDVI to monitor photosynthetic activity. Furthermore, the VCI and TCI are employed to quantify historical vegetation health and thermal stress, respectively, with the VHI providing a synthesised measure of these factors. By merging these variables, the CDI offers a more robust assessment of drought severity than any single index, effectively minimising the noise and uncertainties inherent in single-source monitoring (Liu *et al.*, 2019; Prajapati *et al.*, 2022; Khosh Chehreh and De Michele, 2025).

3.1. Normalisation and scaling

To ensure comparability across multi-source indices with differing units and ranges, all input variables were normalised using min-max scaling (Han *et al.*, 2012). This process rescales values to a uniform range between 0 and 1, a step which is essential for unbiased integration and optimal performance in deep learning architectures. The normalisation is defined by Equation 6:

$$X' = \frac{X - x_{min}}{x_{max} - x_{min}} \qquad \text{Equation 6}$$

Where X norm is the normalised value, X is the original value, X_{min} is the minimum and X_{max} is the maximum, representing the minimum and maximum values of the respective index within the study period.

3.2. CDI weighting using principal component analysis

To find out how each drought indicator contributes to the CDI, we used principal component analysis (PCA). This is a common statistical method for reducing the number of variables (Jolliffe, 2002). To synthesize the multi-scalar information provided by the various drought indices, Principal Component Analysis (PCA) was performed. Instead of treating each index as an independent variable, PCA enabled the identification of Empirical Orthogonal Functions (EOFs), revealing the main spatial patterns and temporal oscillations of drought severity that would otherwise be obscured by localized noise.

The first component explains the most variance in the data. We used the loadings of this component as weights to show the relative importance of each indicator. This ensures the most informative variables have a stronger impact on the CDI (Liu *et al.*, 2019; Prajapati *et al.*, 2022). Before conducting PCA, we normalised all indices to a common scale (Jolliffe, 2002). This kept them comparable and avoided bias from different units or ranges. We calculated the final CDI as a weighted sum of these normalised indices, using weights from the PCA loadings. This approach captures the main patterns of drought variability in the study area. The CDI calculation is based on Equation 7.

$$CDI_t = \sum_{i=1}^{n} w_i \ . \ I_{i,t} \qquad \text{Equation 7}$$

Where $CDI_t$the CDI at time t is, $I_{i,t}$ is the normalised value of the i[th] drought index at time t, and w_i is its corresponding PCA-derived weight. This weighting approach was validated against historical drought records and cross-validation techniques to ensure its reliability for early warning and intervention targeting. While a standardised CDI_t formula was applied across all study areas to ensure methodological consistency, the index was calculated independently for each woreda. This approach captures the distinct spatial and temporal variations inherent to each district, allowing for a comparative analysis of drought intensity across the two regions.

4. Drought classification

After computing the CDI, we categorised the resulting continuous values into discrete severity classes based on SPEI thresholding. This integration leverages the SPEI's standardised benchmarks to ensure the classification remains robust and comparable across diverse environmental contexts. Given that both the CDI and SPEI are derived from similar statistical processes resulting in a Gaussian distribution, they share a common statistical basis and a consistent numerical range. This justifies the application of uniform severity thresholds across both indices (Başakın *et al.*, 2024). By targeting SPEI as a continuous variable rather than discrete classes, the model preserves the full granularity of climatic fluctuations, aligning with international drought monitoring standards (Vicente-Serrano *et al.*, 2010), as detailed in Table 1. This approach facilitates direct comparison with global drought benchmarks while providing a higher resolution of localised environmental impacts.

Table 1: SPEI-based classification of CDI

SPEI range	Drought class
SPEI < -2.00	Extreme drought
-2.00 ≤ SPEI < -1.50	Severe drought
-1.50 ≤ SPEI < -1.00	Moderate drought
-0.99 ≤ SPEI ≤ 0.99	Near normal

5.Model development and evaluation

In this study, we address drought prediction as a regression problem, treating the drought score as a continuous target variable ranging from 0 to 5. To accomplish this, we implement machine learning and deep learning models. The machine learning algorithms included XGBoost, CatBoost, and LightGBM. In the context of drought prediction, these algorithms are categorised as gradient-boosted decision trees (GBDTs), which are particularly effective for drought data as they involve complex, non-linear relationships between weather variables (Bentéjac *et al.*, 2021). The deep learning models include architectures such as bidirectional long short-term memory (Bi-LSTM), CNN-LSTM, and Transformers, selected for their ability to learn from complex spatio-temporal data. Transformer models use self-attention mechanisms to weigh the importance of different input segments, allowing them to capture both spatial and temporal relationships simultaneously (Hochreiter and Schmidhuber, 1997; Han *et al.*, 2012; Niyonsenga *et al.*, 2024). Their ability to efficiently process long-range dependencies offers a distinct advantage over traditional recurrent networks when modelling large-scale climate patterns.

Hybrid models such as CNN-LSTM are also explored, integrating CNN layers for spatial feature extraction with LSTM layers for capturing temporal dependencies. By implementing and evaluating these models using regression-based metrics, we aim to identify the most effective approach for accurate drought intensity prediction across the target regions.

5.1.Machine learning approaches

XGBoost: The XGBoost algorithm functions by building a sequential ensemble of decision trees, where each subsequent tree is specifically trained to minimise the residual errors left by previous ones (Chen et al., 2015). For this study, the XGBoost model is utilised for the regression of drought severity levels, spanning a continuous scale from 0 to 5. To achieve high-performance results, the model's hyperparameters are optimised using a systematic grid search. To enhance generalizability and reduce variance, subsample and colsample_bytree are used to limit the percentage of data points and features considered for each individual tree. Further refinement is achieved through the gamma parameter, which regulates tree growth by requiring a specific threshold of loss reduction for node splits. Additionally, the model applies L1 and L2 regularisation via reg_alpha and reg_lambda to penalise excessive complexity. This comprehensive tuning process, conducted on the validation set, ensures that the hyperparameters (n_estimators=1000, learning_rate=0.05, max_depth=7, Subsample=0.85, colsample_bytree=0.8, Gamma=3, reg_alpha= 0.1, and reg_lambda = 0.1) are calibrated for drought estimation.

CatBoost: CatBoost is a gradient-boosting framework specifically designed to handle categorical features both efficiently and effectively. Unlike other gradient-boosting implementations, CatBoost automatically manages categorical variables by utilising a method known as ordered boosting (Hancock *et al.*, 2020). The CatBoost model hyperparameters are selected through an exhaustive grid search conducted separately for each cluster. Despite the distinct characteristics of each region, the optimal values for all clusters converge to the same set of hyperparameters. The grid search explores variations in iterations, depth, learning_rate, and l2_leaf_reg to balance model complexity and predictive accuracy. The best-performing configuration uses 500 iterations, providing sufficient boosting rounds for learning, and a depth of 7, enabling the model to capture complex patterns without excessive overfitting. A learning_rate of 0.05 ensures stable convergence, while l2_leaf_reg reduces overfitting through L2 regularisation. The fact that the same hyperparameters yield optimal performance across all clusters indicates that CatBoost's learning process generalises well to different regional drought patterns while maintaining high predictive accuracy.

LightGBM: LightGBM is an efficient gradient-boosting framework optimised for speed and performance, especially when dealing with large datasets and high-dimensional data. In contrast to traditional boosting algorithms, LightGBM employs a leaf-wise growth strategy that effectively reduces loss by expanding the most promising leaf nodes (Ke *et al.*, 2017). Key hyperparameters of LightGBM include boosting_type, max_bin, learning_rate, n_estimators, and num_leaves. The boosting_type parameter specifies the boosting method; 'dart' (Dropouts meet Multiple Additive Regression Trees) is used for better generalisation and reduced overfitting. The max_bin parameter (250) controls the maximum number of bins for feature discretisation, with a larger value allowing for finer discretisation at the cost of increased computation. The learning rate is set to a low value (0.05) to ensure gradual learning, while the n_estimators is set to a high level (1,000) to allow the model sufficient iterations to converge. The num_leaves parameter (64) determines the maximum number of leaves in one tree and helps control the model's complexity.

5.2. Deep learning approaches

Bi-LSTM: To capture the cumulative nature of drought development, we implemented a high-capacity BiLSTM architecture capable of modelling long-term temporal dependencies (Zhang *et al.*, 2024). The BiLSTM uses two hidden layers running in opposite directions to capture information from both past and future contexts simultaneously. This mechanism enables the model to selectively retain long-term climatic trends while filtering out transient noise in multi-source input indices (Zhang *et al.*, 2024).

The core of the BiLSTM is the LSTM cell, which manages information flow through a synchronised system of gates. As shown in Equation 8, the forget (f_t), input (i_t), and output (o_t) gates interact to update the cell state (C_t) and produce the hidden state (h_t):

$$\begin{cases} f_t = \sigma\left(W_f.[h_{t-1}] + b_f\right) \\ i_t = \sigma(W_i.[h_{t-1}, x_t] + b_i) \\ \tilde{C}_t = \tanh\,(W_c.[h\_(t-1), x\_t] + \ b_c \\ C_t = f_t * \ C_{t-1} + i_t \ * \ \tilde{C}_t \\ o_t = \sigma(W_0).[h_{t-1}, x_t] + b_0 \\ h_t = o_t * tanh\,(C_t) \end{cases} \quad \text{Equation 8}$$

We structured the input data using a sliding window technique with a 24-month lookback window moved across the dataset with a step size of one. The final configuration employs three LSTM layers with a hidden dimension of 256 units. Optimization Optimisation was performed using the AdamW optimiser, with a learning rate of 2.12 x 10^{-4}, a batch size of 128, and a dropout rate of 0.2 to ensure robust generalisation.

Transformer model: The Transformer model, introduced by Vaswani *et al*. (2017) is an advanced neural network architecture. Unlike traditional recurrent neural networks (RNNs), the Transformer processes input data simultaneously, primarily leveraging self-attention mechanisms to capture relationships and dependencies between all elements of a sequence at once. This mechanism is critical for understanding long-range temporal context in complex climate time series (Han *et al*., 2021).

The main calculation is the scaled dot-product attention, where the attention weights are computed using a query matrix (Q), key matrix (K), and value matrix (V). This operation determines how much focus the model places on other sequence elements when processing a given element, as shown in Equation 9.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad \text{Equation 9}$$

Where d_k is the dimension of the keys.

Our implementation uses a multi-head self-attention approach with a model dimension (d_{model}) of 32 and four attention heads. The architecture includes a single encoder layer with a feed-forward network dimension of 256. Training was performed with a learning rate of 4.07 x 10^{-3}, a weight decay of 8.38 x10^{-5} for L_2 regularisation, and an early stopping patience of 5.

Hybrid models: Hybrid models are machine learning frameworks that combine different architectures to leverage their respective strengths, aiming to improve predictive accuracy and address complex tasks more efficiently than single-model strategies (Huang *et al*., 2024). These models are built on previously optimised base models for each cluster, with hyperparameters carefully selected to maximise performance before integration into hybrid structures (Vo *et al*., 2023).

CNN-LSTM architecture: To leverage both local spatial feature extraction and long-term temporal dependencies, a hybrid CNN-LSTM architecture was developed. The process begins at the input layer, which passes data to the convolutional neural network (CNN) component. This stage uses a 2D convolutional layer (Conv2D) with 32 channels and a kernel size of 5 to identify local patterns and feature correlations within the input data. A subsequent MaxPooling layer reduces dimensionality to emphasise the most significant features.

These filtered outputs are then processed by a LSTM component with a single layer of 128 hidden units, modelling long-term temporal dependencies through its internal memory mechanisms. The resulting sequences are flattened and passed through a series of dense layers. For the regression task, the architecture ends with a fully connected (Dense) layer that outputs the predicted continuous drought index value. This hybrid framework uses a 24-step lookback window and was optimised with the AdamW algorithm at a learning rate of 6.56×10^{-4} and a batch size of 128. To maintain training efficiency, early stopping with a patience of 5 epochs was applied over 60 training iterations. The dropout rate was set to 0, indicating that the data volume was sufficient to balance the architectural complexity without additional stochastic regularisation.

Hyperparameters optimisation: To ensure optimal performance, all deep learning models were rigorously optimised using the AdamW algorithm. The dataset (2000–2022) was partitioned chronologically to prevent data leakage: 60% for training, 20% for validation, and 20% for testing. This split ensures that models are evaluated on unseen future conditions, reflecting a realistic forecasting scenario.

6. Model performance criteria

Model performance criteria are techniques for assessing and comparing machine learning and deep learning algorithms by partitioning the dataset into the following distinct portions: training, validation, and testing (Hastie *et al.*, 2009). The training set is used to build the model, while the validation set serves as a proxy for unseen data to evaluate its effectiveness during development (James, *et al.*, 2021). The specific distribution of data across these three segments is detailed in Table 2.

Table 2: Distribution of the dataset into training, test, and validation sets

Dataset break	Number of observation	Starting date	Ending date
Training data	217,331	2000-03-01	2013-10-01
Validation data	72,444	2013-10-01	2018-05-01
Test data	72,444	018-05-01	2022-11-01

The predictive capability of the selected deep learning models was evaluated using a standard supervised learning protocol. To quantitatively assess prediction accuracy, we used a multi-criteria statistical framework. Although various performance metrics are available in the literature, we utilised a multi-criteria statistical framework to quantitatively assess prediction accuracy. To ensure transparency regarding the 12 month forecasting capability, performance metrics were evaluated based on a specific forecasting lead time (L). Accuracy was measured using the following indicators: the correlation coefficient (R^2) to measure linear association, RMSE to quantify the magnitude of prediction residuals, and MAE to determine the average forecast bias. This combination provides a comprehensive assessment of the models' ability to replicate the complex drought dynamics in Afar and Somali regions.

Correlation coefficient (R^2) is given in Equation 10 (Khan *et al.*, 2020):

$$R^2 = \left(\frac{\sum_{i=1}^{n} (O_i - \bar{O})\,(P_i - \bar{P})}{\sqrt{\sum_{i=1}^{n} (O_i - \bar{O})^2}\,\sqrt{\sum_{i=1}^{n} (P_i - P)^2}} \right)^2 \qquad \text{Equation 10}$$

RMSE is given in Equation 11:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} O_i - Pi)^2}{n}} \qquad \text{Equation 11}$$

MAE is given in Equation 12:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |P_i - O_i \qquad \text{Equation 12}$$

Where n is the number of samples, O_i and P_i denote observed and predicted values, and O⁻ and P⁻ represent the mean of observed and predicted values. Generally, the model with higher R^2 and lower RMSE is considered to be the more accurate model. This approach determines the performance of each model in continuous time series estimation and ensures that the drought monitoring capability is not biased toward specific high-density station locations.

7. Afar Region exploratory data analysis

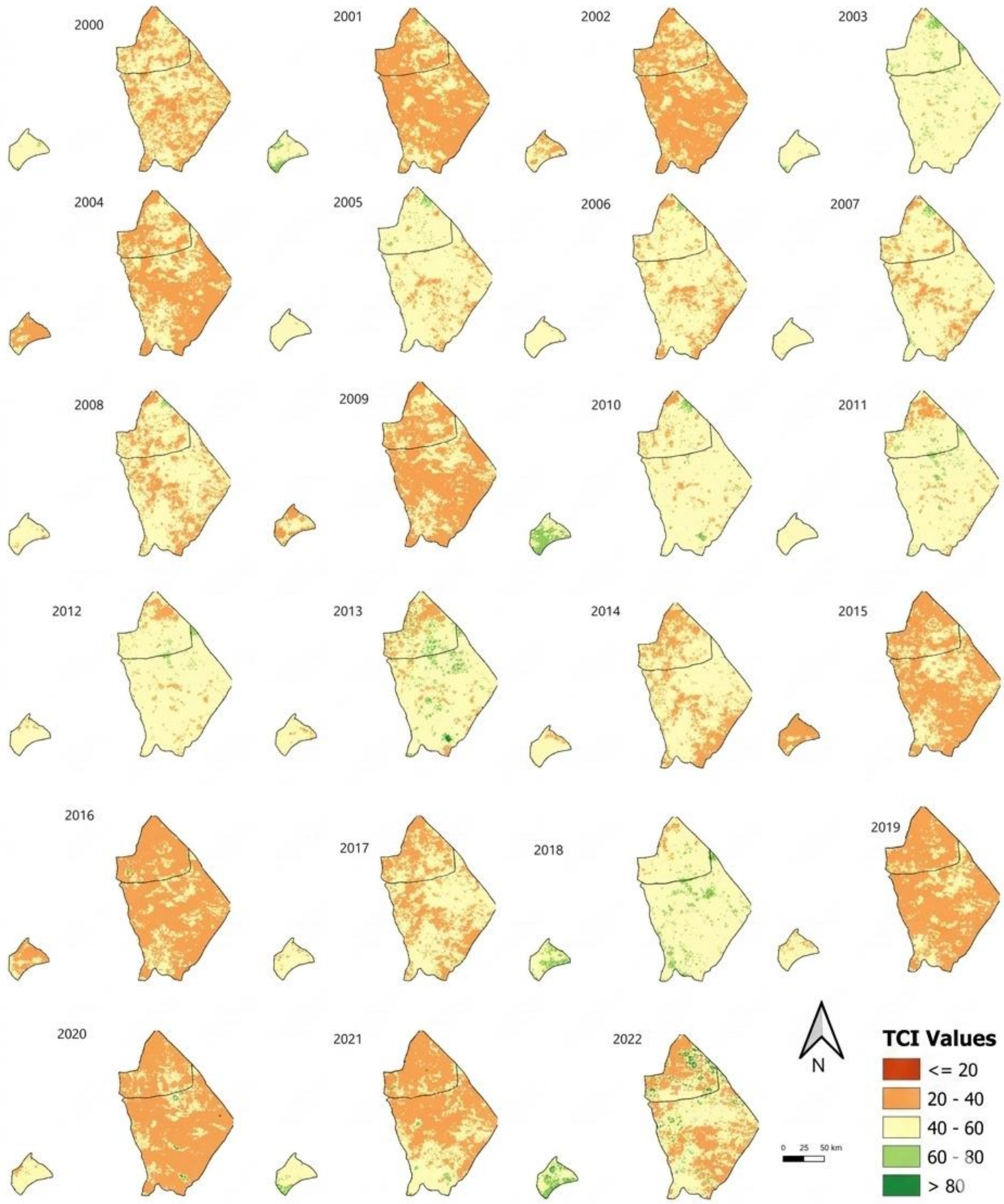
7.1. TCI

Our analysis of the TCI for the Karma growing season (July–September) in the years 2000–2022 reveals consistent temperature-induced vegetation stress in the Bidu, Elidar, and Ewa woredas, see Table 3. By calculating the drought index exclusively for the Karma season, the model isolates the primary moisture-stress window and avoids dilution from off-season data. This is critical, as high thermal stress during this period accelerates evapotranspiration and exacerbates moisture deficits. As summarised in Table 3 and Figure 3, TCI values frequently fell below 40 (Kogan, 1995b), indicating widespread severe drought, particularly in Elidar and Bidu. These two woredas experienced compounding chronic stress, most notably during the 2019–2021 multi-year anomaly while optimal conditions (TCI > 80) were largely absent. In contrast, Ewa exhibited greater resilience. This is likely due to its higher average elevation (approximately 1,200 metres, compared to 600 metres in Bidu), which, due to orographic effects, results in a consistently lower LST throughout the growing season. This temperature differential appears to modulate thermal stress and reduce its impact on local vegetation (Burka *et al*., 2024). These findings highlight the need to integrate TCI-derived thermal stress data with the complementary precipitation and vegetation indices discussed in the following sections to establish a robust, multi-dimensional drought monitoring system.

Table 3: Summary of rainfall patterns and TCI-derived drought years (2000–2022)

Woreda	Rainfall pattern (Karma focus)	Severe drought years (TCI < 40)	Favourable/better years (TCI≥ 40)	Observations
Elidar	Unimodal: highly arid; >80% of annual rain occurs in July–Sept.	2001,2002,2009, 2016,2019–2021	2005, 2007, 2010, 2012, 2014, 2018	Severe thermal stress dominates; optimal conditions (TCI > 80) were absent.
Bidu	Unimodal: concentrated Karma rains; Belg (short rains) almost absent.	2001, 2002, 2009, 2016, 2019–2021	2005, 2007, 2010, 2012, 2014, 2018	Mirroring Elidar's pattern; extreme aridity limits recovery periods.
Ewa	Unimodal: main rains July–Sept; minimal/negligible Belg.	2004, 2015, 2016	2001, 2010, 2018, 2022	Higher resilience; several no drought years (TCI > 80) recorded.

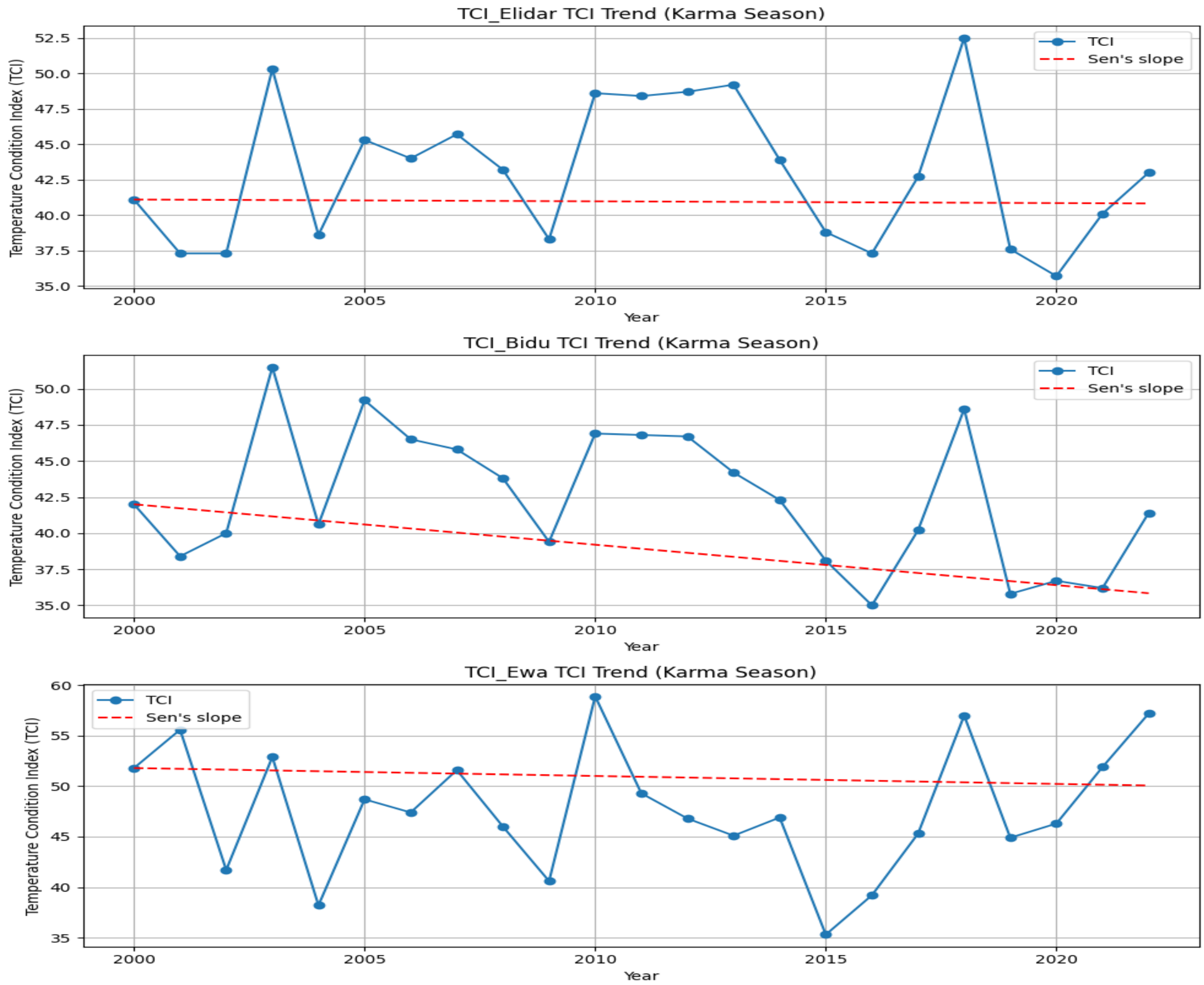
Figure 3: Spatio-temporal distribution of the TCI for the Karma season



7.2. Mann-Kendall test and Theil-Sen trend analysis

The Mann-Kendall test, combined with Theil-Sen trend analysis, was applied to evaluate the drought changes in Bidu, Elidar, and Ewa woredas during the Karma season in the years 2000 to 2022 (Mann, 1945; Sen, 1968). The results indicate that, overall, the temperature-induced vegetation stress, as represented by TCI, shows no significant monotonic trend in the study area, see Figure 4. The Theil-Sen slope values range from - 0.28 to - 0.01 TCI units per year, reflecting minimal overall change in drought severity during the observed period. However, localised variability exists within the region. Bidu exhibits a weak negative trend, suggesting a slight increase in drought stress, though this is not statistically significant, while Ewa and Elidar demonstrate stable conditions with near-zero slopes. This pattern indicates that while broad-scale drought severity remains largely unchanged during the Karma season, some areas may experience subtle shifts in drought dynamics.

Figure 4: Inter-annual TCI trends and Theil-Sen slope estimates for the Karma season (2000–2022) in Elidar, Bidu, and Ewa woredas



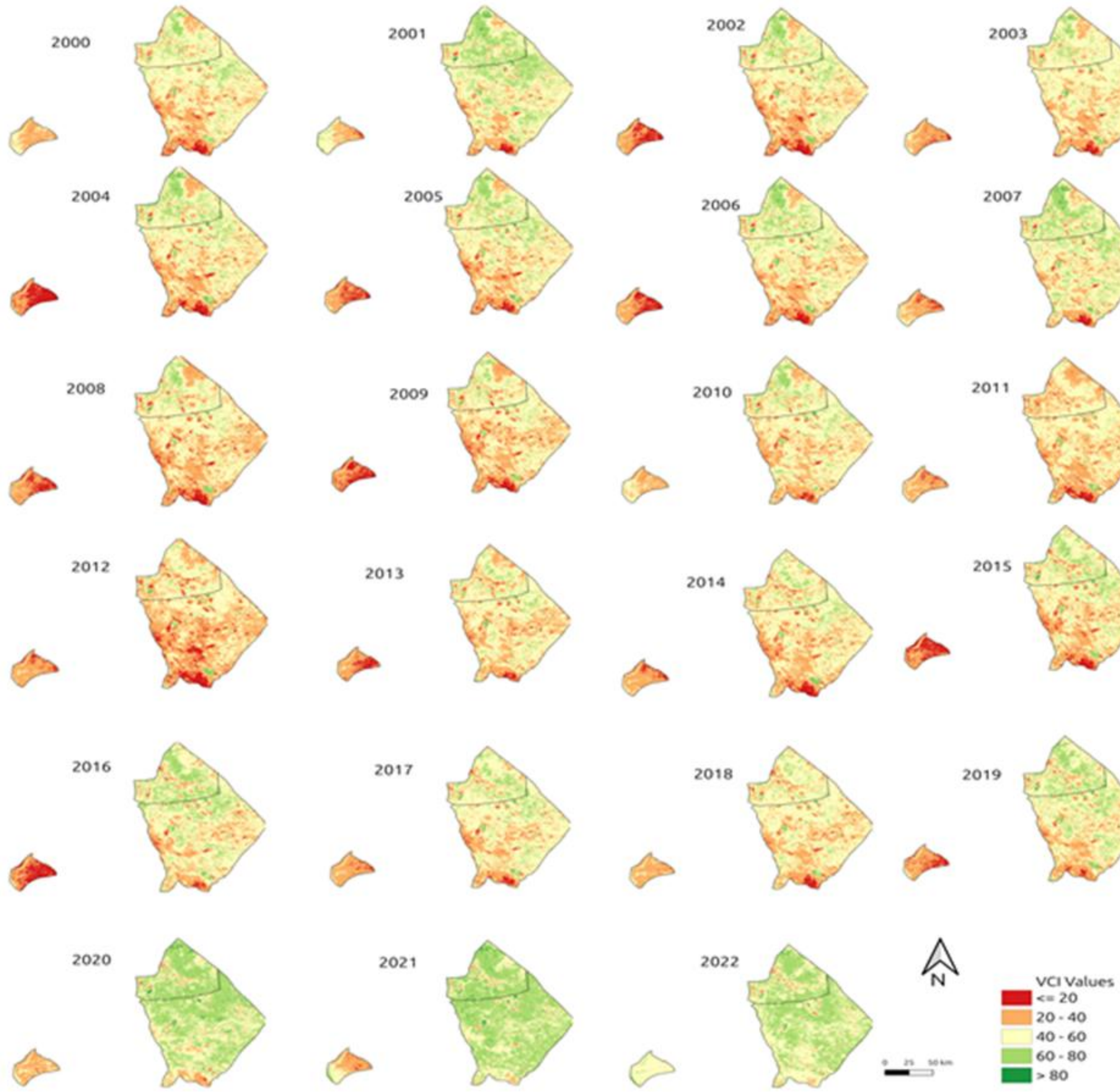
These findings highlight the importance of continuing drought monitoring with complementary indices and finer spatial analyses to capture localised drought intensification or relief within the arid and semi-arid Afar Region. The results show no statistically significant monotonic trends in drought severity overall, with Theil-Sen slopes indicating minimal change. Localised variability suggests subtle shifts in temperature-induced vegetation stress, emphasising the need for continued drought monitoring in the region.

7.3. VCI

The VCI was analysed for Bidu (northeast), Elidar (southeast), and Ewa (west) woredas in Afar Region using a five-class scheme (<20%, 20–40%, 40–60%, 60–80%, and >80%) (Kogan, 1995b) to evaluate spatial and temporal drought patterns, see Figure 5. Analysis showed that Bidu and Elidar are persistent drought hotspots. Bidu experienced predominantly extreme to severe conditions, with large areas maintaining VCI values below 40%, especially during severe years, such as 2002, 2011, and 2015. Elidar also experienced frequent severe to moderate drought (VCI 20–60%), with notable severe conditions in 2009, 2011, and 2017. In both woredas, recovery phases were brief, though relatively better vegetation conditions (VCI >60%) occurred in 2001, 2007, and 2020–2022, reflecting favourable rainfall patterns.

In contrast, Ewa showed significantly greater resilience, with more frequent occurrences in the mild drought to favourable range. Higher and more stable vegetation health was recorded in Ewa during wetter years, such as 2006, 2010, and 2018. These spatial patterns align with known arid climatic zones and previous drought assessments in the region. The contrast between the chronic vulnerability of the eastern lowlands (Bidu and Elidar) and the better agro-ecological recovery in the west (Ewa) highlights the critical need for localised drought monitoring and targeted mitigation strategies across Afar Region.

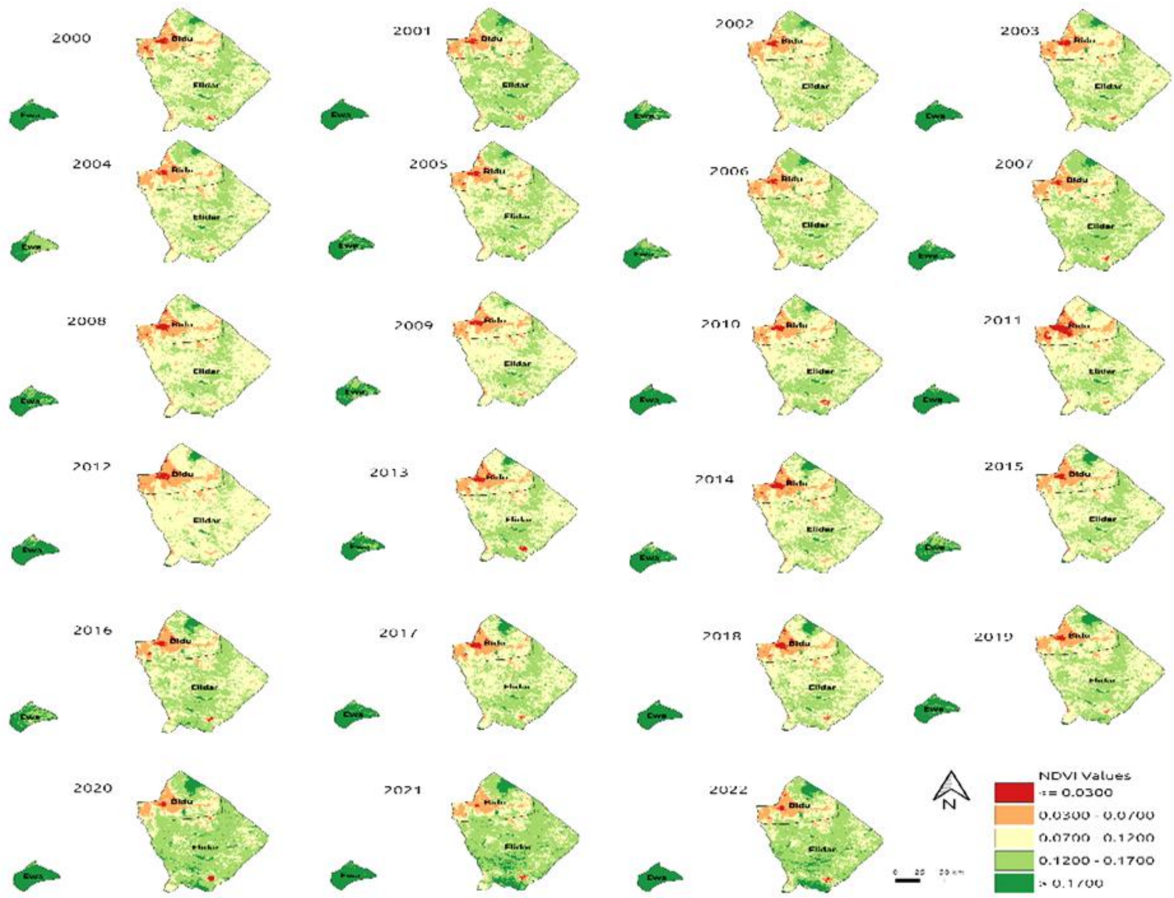
Figure 5: Spatial distribution and inter-annual variation of the VCI during the Karma season (2000–2022)



7.4. NDVI

Spatially, a persistent north–south gradient is evident, with the northern woredas (Bidu and Elidar) consistently exhibiting low NDVI values (≤0.0700) (Tucker, 1979) (see Figure 6), indicative of sparse vegetation and high aridity, while Ewa in the south maintains higher NDVI values (> 0.1700), reflecting more favourable vegetation conditions (Tucker, 1979). Temporally, pronounced vegetation stress is observed during 2009, 2015, and 2017, coinciding with known regional drought episodes (U.S. Geological Survey, 2018). From 2019 onwards, a gradual recovery is evident, culminating in widespread green cover in 2021–2022, particularly in Elidar and Ewa, suggesting improved rainfall conditions or localised vegetation restoration. These patterns highlight both northern Afar's chronic vulnerability to drought and the relative ecological resilience of southern zones.

Figure 6: Spatio-temporal distribution of NDVI in Afar Region (2000–2022)

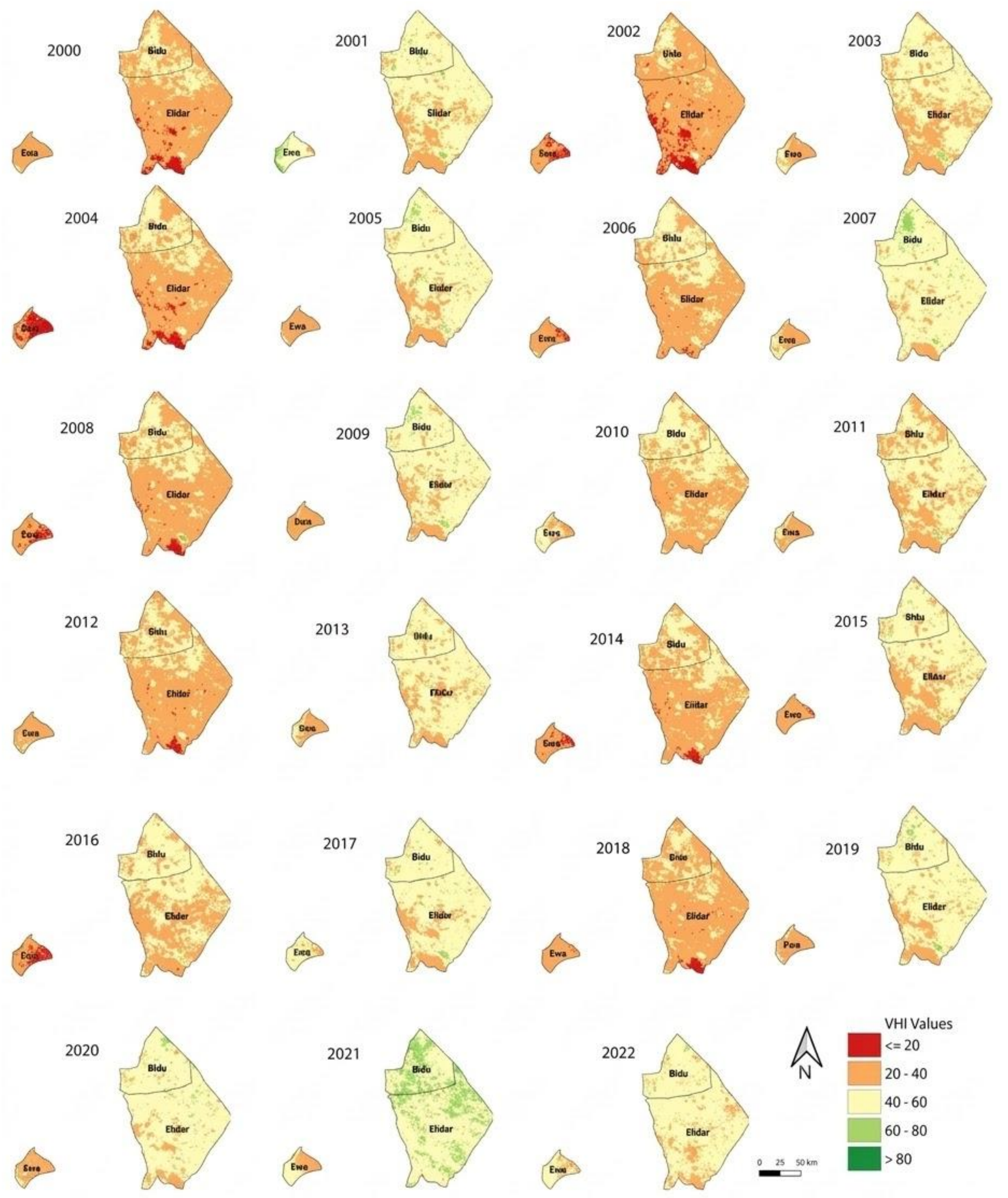


7.5. VHI

The temporal analysis of the VHI reveals a landscape characterised by high climatic variability and recurring environmental stress (Abrahim *et al*., 2025), particularly in Elidar and Ewa districts, see Figure 7. Between 2000 and 2022, the region experienced frequent cycles of severe to extreme drought (VHI ≤ 40) (Zeng *et al.*, 2023; Alahacoon and Edirisinghe, 2022; Kogan, 1995c), with the most critical periods in 2002, 2004, 2012, and 2018. During these years, extensive portions of the southern study area fell into the extreme drought category (VHI ≥ 20), indicating acute moisture deficits and near-total vegetation failure. These patterns highlight persistent vulnerability to climatic shocks, with the southern regions serving as primary indicators of environmental degradation during dry spells.

In contrast, Bidu district in the north demonstrates greater ecological resilience, often maintaining fair to good vegetation health even when southern areas are heavily stressed. This spatial disparity is most evident during recovery years such as 2007 and 2021, when VHI values in Bidu and northern Elidar exceeded 60, reflecting robust biomass production. Despite these periods of recovery, Ewa district remains a consistent drought hotspot, frequently remaining in the 20–40 VHI range even during relatively stable years. This suggests that while the broader ecosystem is capable of rapid regeneration following rainfall, the localised sensitivity of Ewa and southern Elidar requires targeted climate adaptation strategies to mitigate the impact of recurrent drought.

Figure 7: Spatio-temporal distribution of VHI for Bidu, Elidar, and Ewa districts from 2000 to 2022

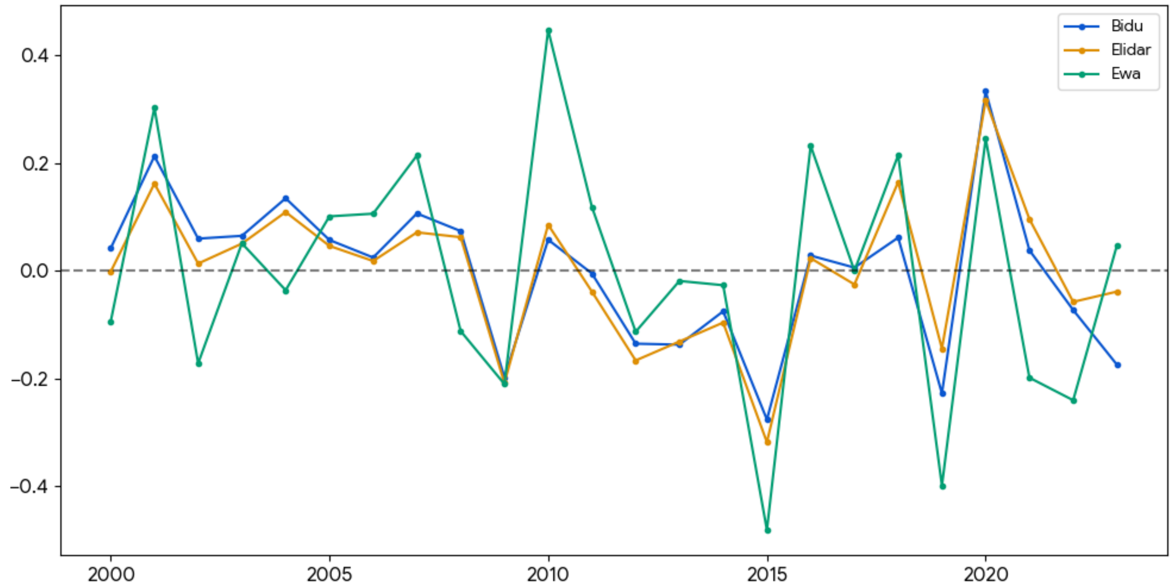


7.6. SPEI

The SPEI provides a critical climatic baseline for understanding the net moisture balance in the Afar Region, integrating both precipitation and potential evapotranspiration to capture environmental water stress (Oksal, 2023). As shown in Figure 8, the region is marked by high inter-annual variability, with a significant drought frequency of approximately 0.192. Contrary to studies suggesting a permanent shift towards wetter conditions, our findings indicate that the climate remains highly volatile.

From 2018 to 2021, there were notable peaks in moisture, with 2020 recording the wettest conditions on record. However, this was immediately followed by a return to severe drought in 2022. Within the historical record, 2015 and 2019 stand out for their widespread moisture deficits (SPEI ≤ -1.5). The peak intensities of these events were highly localized: From 2018 to 2021, there were notable peaks in moisture, with 2020 recording the wettest conditions on record. However, this was immediately followed by a return to severe drought in 2022. Within the historical record, 2015 and 2019 stand out for their widespread moisture deficits (SPEI ≤ -1.5). The peak intensities of these events were highly localized: during the 2015 event, Ewa recorded an extreme value of -2.43, while Bidu and Elidar registered -1.21 and -1.18, respectively. During the 2019 event, Bidu experienced a significant drought at -1.93, while Ewa and Elidar recorded more moderate conditions at -0.87 and -1.02, respectively.

Figure 8: Yearly average SPEI by selected woredas: Ewa, Bidu, and Elidar woredas (2000–2023)



In summary, integrating TCI, VCI, NDVI, VHI, and SPEI datasets shows that drought dynamics in Afar Region are shaped by a complex interplay of elevation, thermal regulation, and bimodal moisture availability. Bidu and Elidar remain the most vulnerable zones, exhibiting synchronised susceptibility to chronic aridity, while Ewa stands out as a resilient outlier. Although there is no statistically significant long-term worsening trend, the recurrent mid-series depressions identified between 2009 and 2015 underscore the region's precarious equilibrium. These findings highlight that effective drought mitigation must go beyond broad regional generalisations and instead focus on localised interventions that consider the specific thermal and vegetative thresholds of each district.

8. Somali Region exploratory data analysis

8.1. TCI

The TCI exploratory analysis for Somali Region (2000–2022) reveals a landscape characterised by intense inter-annual volatility and sharp spatial contrasts. In this region, thermal stress serves as a high-resolution indicator of bimodal rainfall performance. Unlike the relatively stable chronic stress observed in Afar Region, the Somali woredas experience acute thermal shocks that are closely aligned with the timing of the Gu (March–May) and Deyr (October–December) seasons (Gebremeskel *et al.*, 2025).

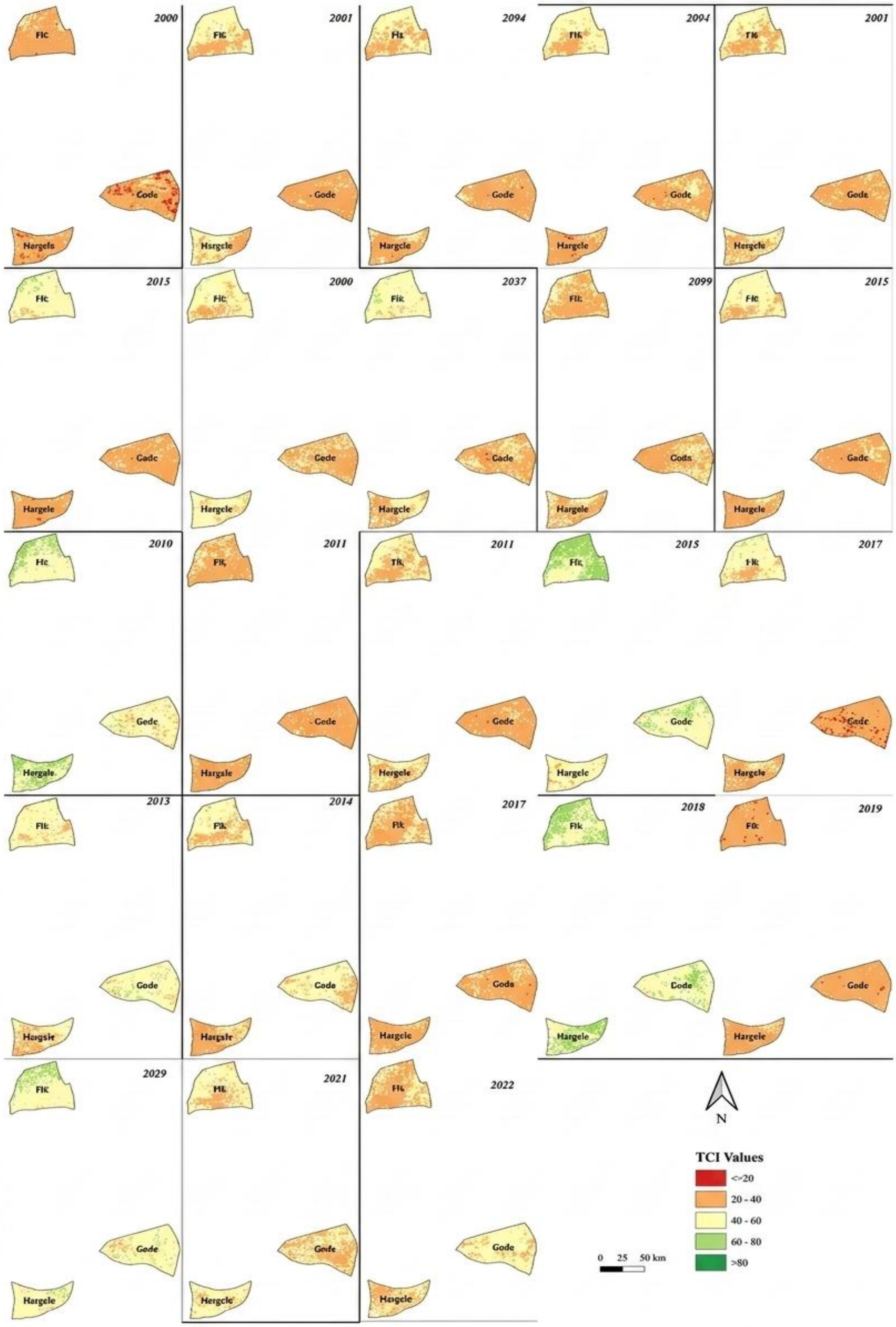
In this ecosystem, the absence of early-season rainfall triggers a rapid increase in sensible heat flux and a loss of evaporative cooling, resulting in the severe TCI depressions identified in the study. Specifically, Gode (central Somali) emerges as a persistent thermal hotspot, frequently recording extreme heat stress (TCI < 10), likely exacerbated by its low-lying topography within the Shabelle River basin. In contrast, Fik and Hargele demonstrate greater periodic resilience: favourable years such as 2002, 2012, and 2017 were characterised by reduced temperature stress and moderate TCI values (above 40).

In contrast, extreme drought years marked by regional rainfall failure led to widespread thermal intensification across all sites, reinforcing the region's susceptibility to climatic volatility. As shown in Table 4 and Figure 9, the timing of these thermal vulnerabilities is determined by geographical variations in rainfall. Northern woredas, such as Fik, experience strong Gu and Deyr influences, with a significant dry summer (Hagaa) from June to September. In the southwestern areas near Hargele, the bimodal peaks are more pronounced, with a shorter intervening dry period. Gode typically experiences a dominant Gu season, while the Deyr peak is often weaker and more variable between years, making the area more vulnerable to late-year thermal stress.

Table 4: Rainfall patterns in selected Somali Region woredas

Woreda	Location context	Rainfall pattern	Notes
Fik	Northern Somali	Bimodal	Strong Gu (Mar–May) and Deyr (Oct–Dec) seasons; dry summer (Jun–Sep).
Hargele	Southwestern Somali	Bimodal	Pronounced Gu and Deyr rains, with a short dry season between peaks.
Gode	Central Somali (Shabelle River)	Bimodal (but weaker second peak in some years)	Gu is generally the stronger peak; Deyr is significant but can be weaker in some years.

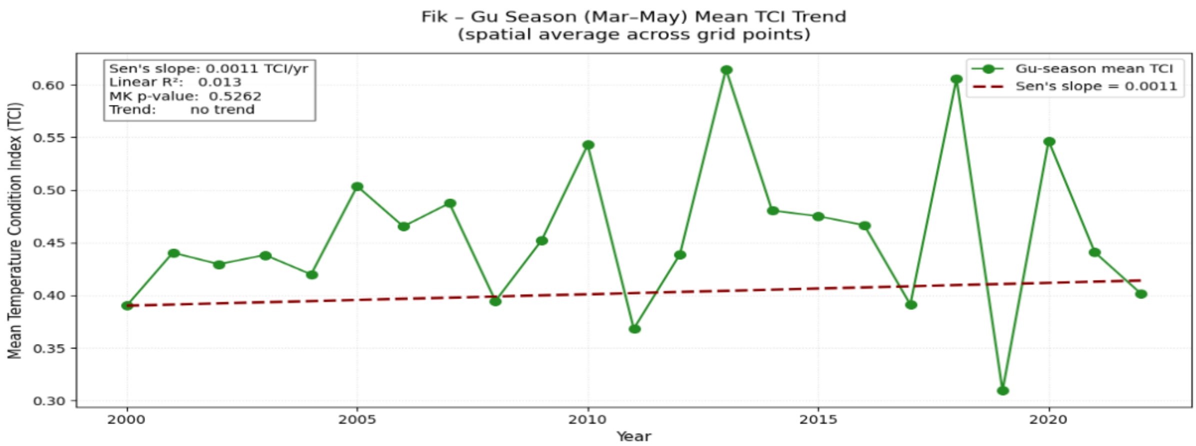
Figure 9: Spatio-temporal variation of the TCI in Somali Region (Fik, Gode, and Hargele), Ethiopia, 2000–2022

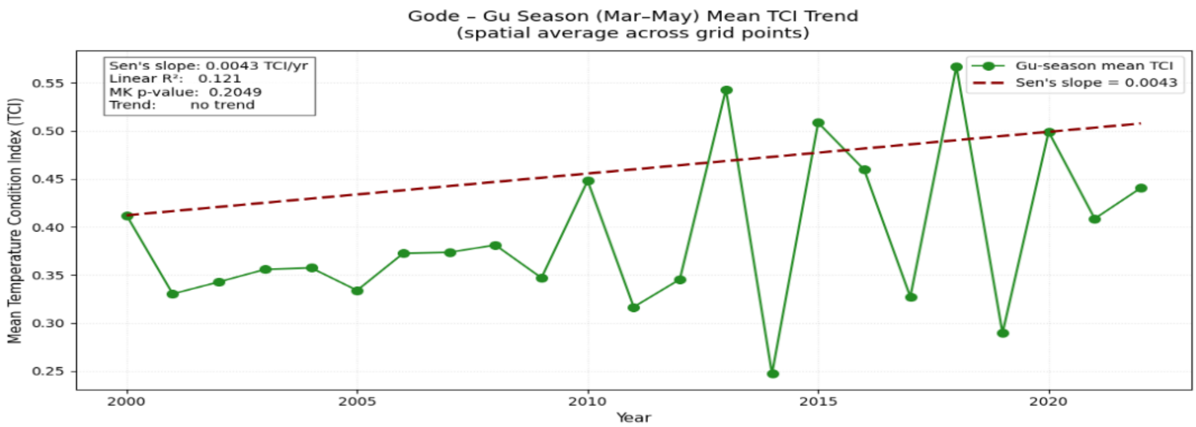


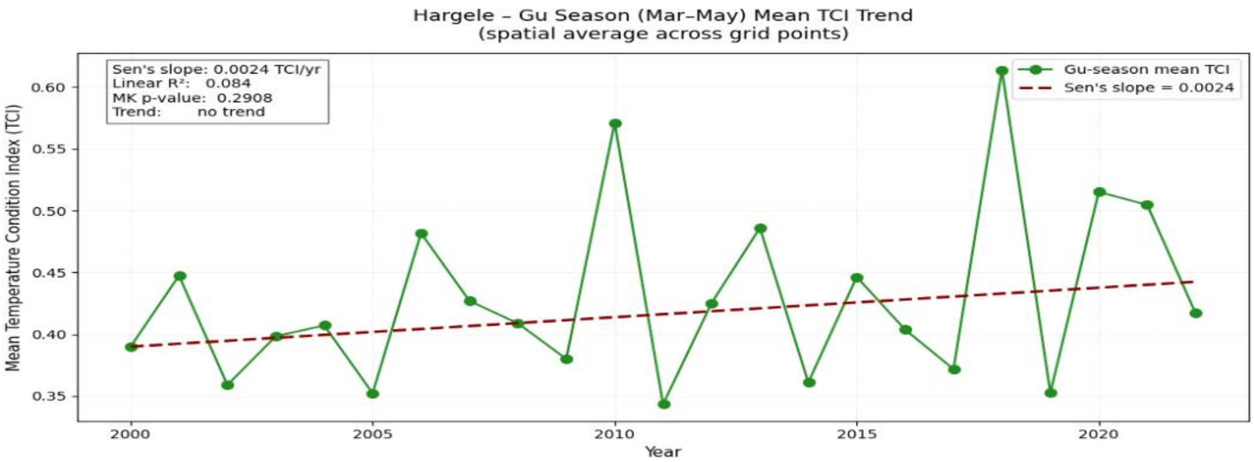
8.2. Mann-Kendall test and Theil-Sen trend analysis

A Mann-Kendall trend test and Theil-Sen slope estimator were applied to thermal data from Somali Region (2000–2022) to assess long-term changes in climatic pressure, see Figure 10. Given the region's bimodal rainfall pattern, the analysis focused on the Gu (March–May) season, which is the main growing period, to ensure that long-term climatic signals were not masked by intervening dry seasons. The results reveal a complex thermal trajectory that contrasts sharply with the stationary stress observed in Afar Region. Most notably, Gode exhibited a highly significant positive trend ($p = 0.0011$; Sen's Slope: +0.0128), indicating a gradual reduction in thermal stress intensity during the spring months over the 22-year period. Although Gode remains the hottest woreda in absolute terms (TCI < 10), this upward trend suggests a localised moderation of extreme heat shocks, potentially related to shifting moisture patterns or increased cloud cover. In contrast, Fik exhibited a stationary trend ($p = 0.4596$), with a negligible slope (+0.0017), indicating that despite high inter-annual volatility, the woreda's baseline thermal environment has not experienced a progressive monotonic change. Similarly, Hargele showed no significant trend, although the analysis was limited by greater data fragmentation, typically due to cloud interference during the bimodal peaks. Together, these findings indicate two distinct climate change signatures: northern Afar Region experiences stationary chronic stress, while Somali Region, particularly Gode, is characterised by volatile recovery, with extreme thermal peaks increasingly moderated by a significant, though gradual, cooling trend during the primary growing season.

Figure 10: Trends in thermal stress (TCI) during the Gu season for selected woredas in Somali Region (2000–2022)







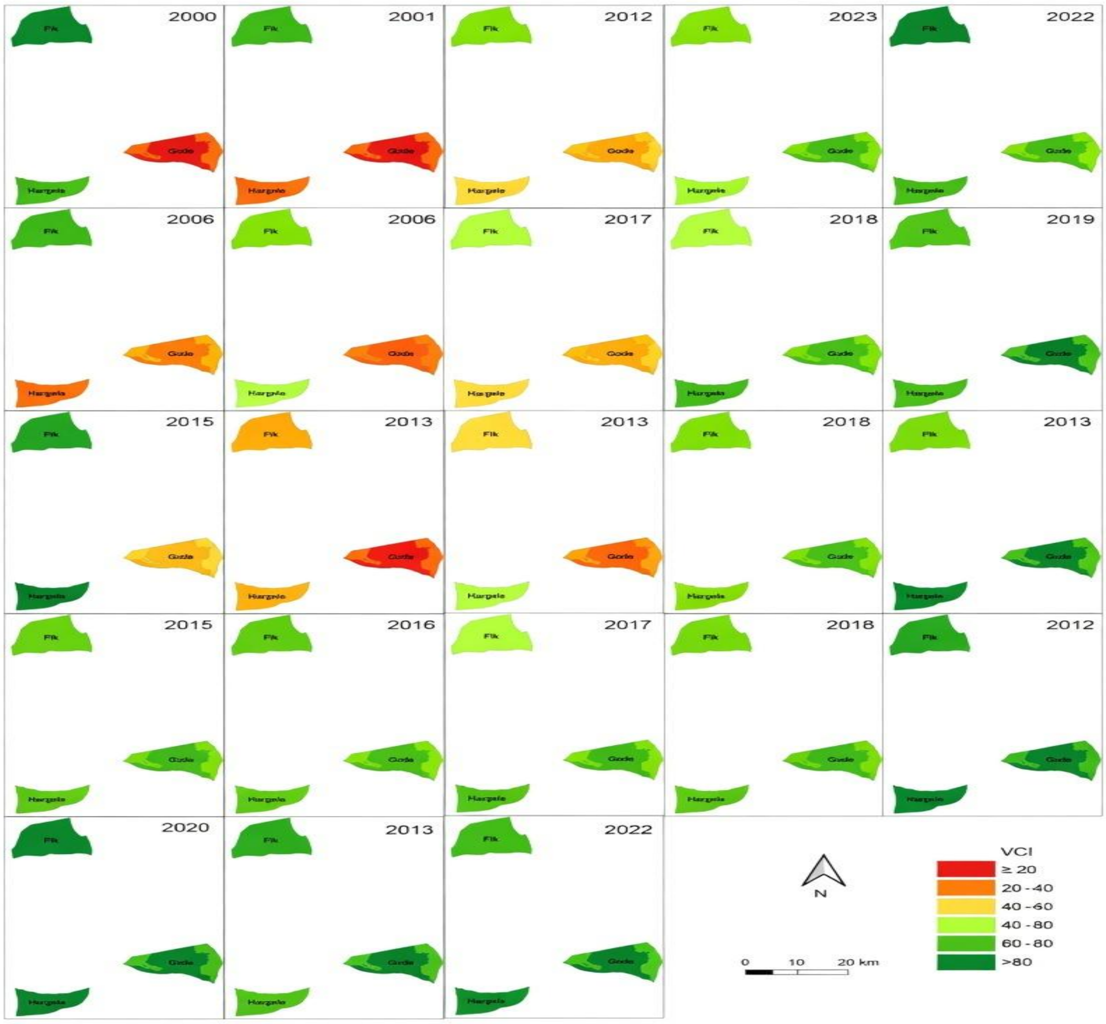
8.3. VCI

The exploratory analysis of the VCI for Somali Region (2000–2022) provides a high-resolution view of rangeland productivity and biomass deficits during the primary Gu (April–June) growing season (see Figure 11). As a normalised measure of current NDVI relative to its long-term historical range, the VCI serves as a direct proxy for the biophysical impact of the climatic stressors discussed in the sections detailing the TCI and rainfall results.

The analysis reveals intense inter-annual variability, with acute vegetation stress indicated by widespread red and orange dominance (VCI < 35) during the major drought episodes of 2000, 2005, 2011, 2015, and 2017. These periods represent regional collapses in rangeland health are that correlated directly with the thermal shocks identified in the TCI analysis. In contrast, significant recovery years, such as 2006, 2010, and 2012-display robust green dominance (VCI > 50), highlighting the remarkable, though fragile, resilience of Somali pastoralist ecosystems following favourable rainfall onsets.

Spatial heterogeneity remains a defining feature of the region's drought dynamics. Gode consistently emerges as the most chronically stressed woreda, reflecting the extreme thermal pressure and significant TCI trends observed in the central Shabelle basin. While Fik and Hargele show more frequent localised recovery, recent data from 2020–2022 indicate persistent and intensifying stress across all sites. This downturn suggests that the cumulative impact of consecutive failed rainy seasons is overwhelming the region's natural recovery mechanisms, significantly increasing livelihood risks for pastoralist communities that are dependent on these rangelands.

Figure 11: Spatio-temporal distribution of the VCI for Somali Region during the Gu season (2000–2022)



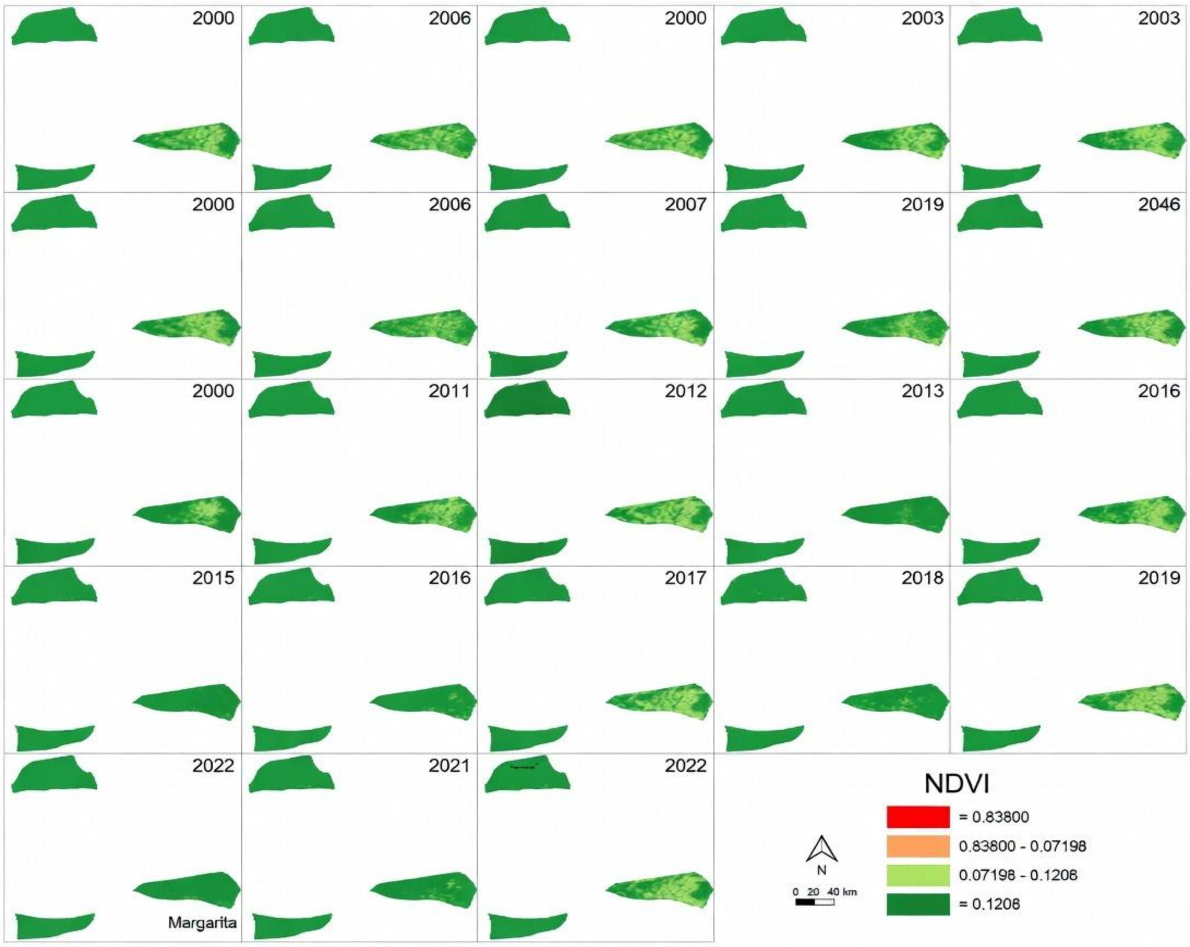
8.4. NDVI for the Gu season

The multi-temporal NDVI analysis for the Gu season (2000–2022) across Fik, Gode, and Hargele provides a quantitative record of absolute vegetation vitality and photosynthetic activity, as shown in Figure 12. As a primary indicator of live green biomass, NDVI reflects vegetation condition, which forms the biological basis for the previously discussed VCI trends. The results show a high degree of synchronisation between greenness levels and bimodal rainfall patterns, capturing the rapid transition of pastoral landscapes from seasonal dormancy to peak productivity.

The satellite time series highlights marked inter-annual variability. Years such as 2002–2004, 2013-2014, and 2019 showed relatively high NDVI values (indicated by deep green hues in the imagery), signalling favourable rainfall and strong biomass accumulation. In contrast, acute drought years-most notably 2005-2006, 2010-2011, and 2015-2017-were characterised by widespread declines in NDVI.

The analysis reveals distinct regional resilience profiles. Fik showed more consistent greenness over the 22-year period, suggesting a higher ecological buffer or better soil moisture retention than the southern woredas, Gode and Hargele. In contrast, Gode exhibited the most pronounced fluctuations, swinging sharply between peak greenness and extreme browning. Hargele generally displayed lower baseline vegetation vigour, making it particularly susceptible to even minor rainfall delays. These NDVI dynamics highlight the precarious nature of the Somali ecosystem, where recurrent drought-induced stress directly undermines the sustainability of pastoral livelihoods.

Figure 12: Spatio-temporal dynamics of vegetation greenness (NDVI) in Somali Region during the Gu season (2000–2022)

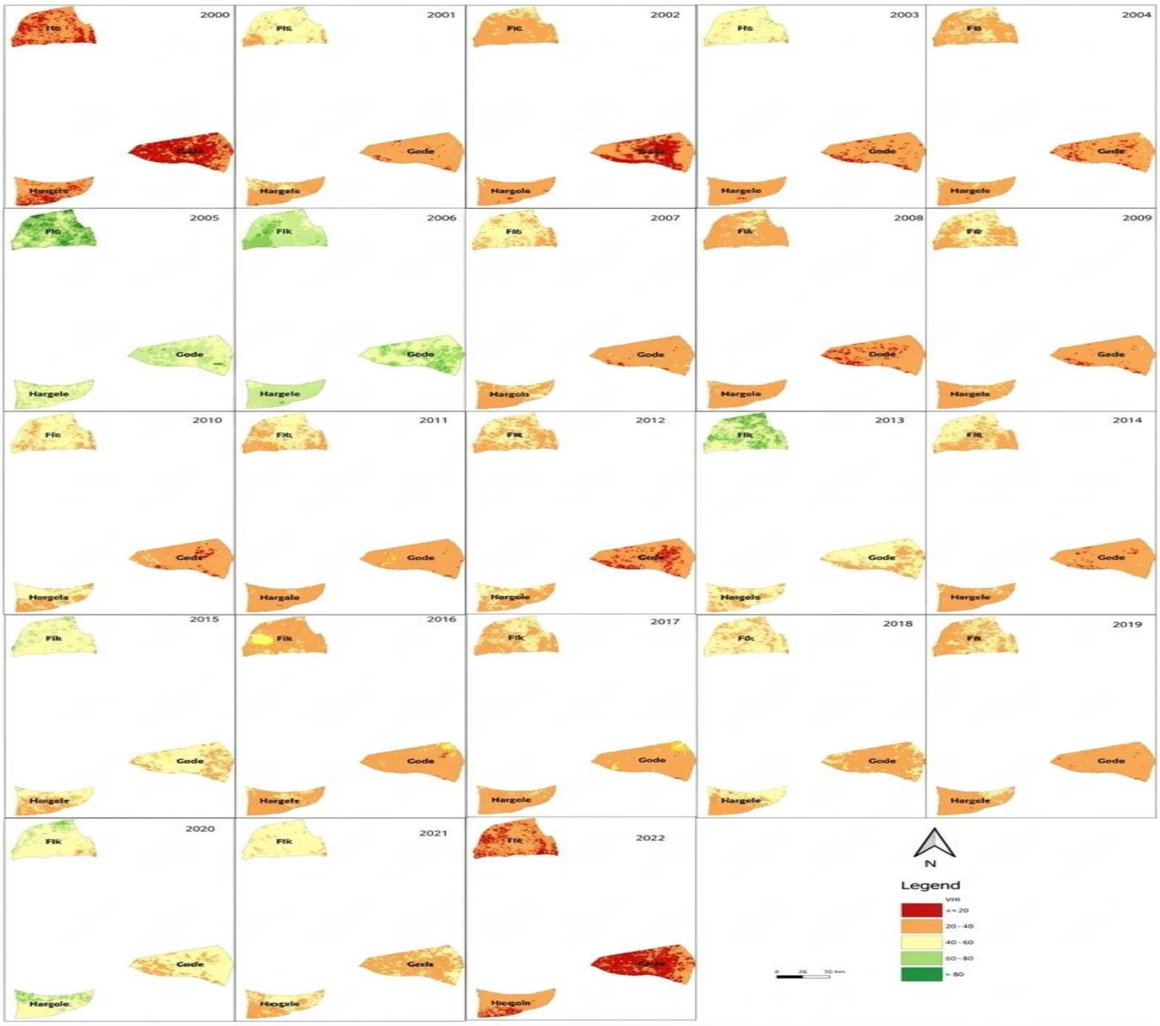


8.5. VHI

The spatial analysis of the VHI from 2000 to 2022 confirms that Fik, Gode, and Hargele are increasingly exposed to extreme environmental volatility, with drought becoming the prevailing climatic condition rather than an occasional anomaly. The data show a clear cyclical pattern of vegetation failure, with 2022 marking the most severe ecological collapse in the 23–year period for all three study areas, see Figure 13. Gode consistently remains the most vulnerable district due to chronic thermal and moisture stress, but the rapid decline in

Fik and Hargele during 2021–2022 indicates a regional homogenisation of drought impact. These findings highlight that the shrinking windows for ecosystem recovery are inadequate for land regeneration, posing a critical threat to local livelihoods. This longitudinal study therefore emphasises the urgent need to integrate high-resolution satellite indices into early warning systems to mitigate the socio-economic impacts of intensifying drought cycles in Somali Region.

Figure 13: Spatio-temporal distribution of the VHI in Fik, Gode, and Hargele districts from 2000 to 2022



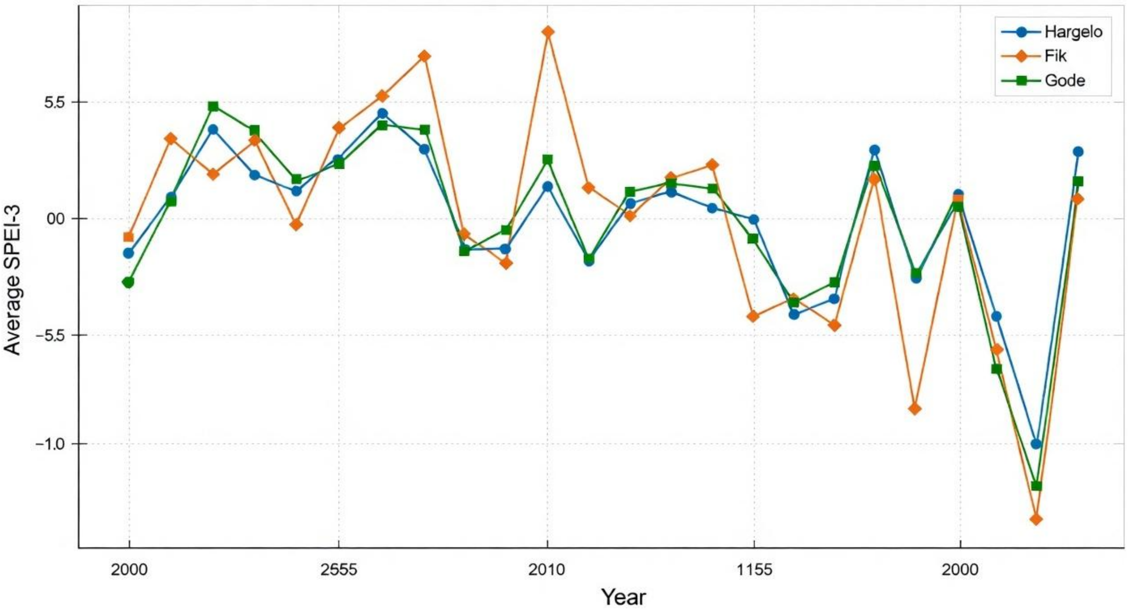
8.6. The SPEI

The analysis of the SPEI-3 dataset for the Fik, Gode, and Hargele woredas from 2000 to 2023 reveals a region defined by high moisture variability and persistent drought cycles, see Figure 14. Utilising K-means clustering to delineate the three woredas across 513 grid locations, the data indicate that approximately 0.147 of the months in Hargele reached drought thresholds (SPEI-3 ≤ -1.0), making it the most vulnerable of the three study areas. Across the entire region, nearly 0.07 of all months recorded severe to extreme drought conditions (SPEI-3 ≤ -1.5). The year 2022 stands out as the most severe drought year in the

23-year period, characterised by a prolonged moisture deficit that surpassed previous historical lows in 2012 and 2019.

Beyond this inter-annual variability, the region exhibits a bimodal moisture pattern at the seasonal scale, where the Gu rainy season (March - May) shows a critical vulnerability; March is consistently the driest month, frequently failing to provide the necessary moisture for early-season planting or rangeland rejuvenation. Conversely, the Deyr season (October–December) serves as the primary period of moisture recovery, with November and December showing the highest average SPEI values. These findings underscore the recurring nature of climatological stress in Somali Region and highlight the increasing intensity of drought events in the last decade, pointing to the need for robust climate adaptation strategies for the local pastoralist and agro-pastoralist communities.

Figure 14: Yearly average SPEI-3 by selected woredas: Fik, Gode, and Hargele woredas (2000–2022)



The Exploratory Data Analysis (EDA) in this study integrates five distinct environmental indices - SPEI, TCI, NDVI, VHI and VCI to construct a comprehensive account of the region's ecological and climatic health. NDVI and VCI are the primary analytical tools for monitoring biomass and relative plant health, effectively capturing the physical greening during periods of high moisture and the subsequent browning during drier periods.These biological responses are fundamentally driven by fluctuations in net moisture balance (monitored via SPEI) and elevated thermal stress (identified through TCI), which together accelerate moisture loss from soil and vegetation. By synthesising these metrics, the EDA reveals a pattern of climatic fragility, characterised by temporary pulses of vegetation growth frequently interrupted by sharp, deep-seated drought cycles.

This multi-dimensional approach demonstrates that, while surface greenness may occasionally rebound, persistent water deficits and extreme atmospheric heat continue to threaten the long-term stability of rangeland management and food security. These dynamics are particularly evident across the study's focal areas, including Ewa, Elidar, and Bidu in the Afar Region, as well as Fik, Gode, and Hargele in the Somali Region.

9. Model performance evaluation

To identify the most precise forecasting method, we benchmarked three machine learning and three deep learning models using data from the six selected woredas. The deep learning architectures generally outperformed the machine learning models (as shown in Table 5), mainly due to their intrinsic capacity to model complex spatio-temporal dependencies. Given that drought patterns in the selected woredas are inherently tied to spatial disparities, seasonal variations, and temporal shifts, the selected models are integrated into these multi-dimensional variables (Chandra et al., 2025).

To evaluate the predictive accuracy and robustness of these models, this study uses two distinct climate metrics, i.e. the SPEI and the CDI. These indices were selected mainly for their proven sensitivity to hydrological fluctuations in arid, and semi-arid landscapes. Previous studies suggest that while traditional metrics may overlook minor thermal shifts, the SPEI and CDI offer better performance in predicting drought severity within semi-arid landscapes, where evapotranspiration plays a critical role in moisture deficits (Vicente-Serrano *et al.*, 2010; Bozorgi *et al.*, 2025).

We evaluated the six models to identify the most effective drought forecasting tool for the arid and semi-arid landscapes of Afar and Somali regions. Table 5 presents a comparative analysis of the models trained on the SPEI. Model performance was measured using three key metrics: MAE, RMSE, and the coefficient of determination (R2). Accordingly, the CNN-LSTM model (MAE = 0.27505, RMSE = 0.37884, R2 = 0.82492) generally outperformed both the Bi-LSTM and Transformer-based approaches. This stronger performance is attributed to the model's hybrid architecture, which leverages convolutional layers to extract local spatial features, while utilising the LSTM component to capture long-term temporal dependencies (Zhang *et al.*, 2024; Vo *et al.*, 2023).

Table 5: Performance evaluation of models using the SPEI

Model	MAE	RMSE	R2
XGBoost	0.4377	0.5984	0.5816
LightGBM	0.4286	0.5839	0.6016
CatBoost	0.4160	0.5678	0.6232
CNN-LSTM	**0.27505**	**0.37884**	**0.82492**
Bi-LSTM	0.71372	0.71582	0.47829
Transformer-based model	1.34289	1.56121	0.25487

Beyond a single-index analysis, we conducted experiments using a CDI, as shown in Table 6. This composite metric integrates several remote sensing and climatic indicators, including NDVI, VCI, TCI, VHI, and SPEI, to provide a holistic measurement of drought severity. Table 6 summarises the performance of the six machine learning architectures for this multivariate configuration. This approach evaluates the models' ability to predict an aggregate signal that captures the complex interactions between vegetation health, thermal stress, and water deficits (Lim and Zohren, 2021).

Table 6: Model performance comparison for CDI regression

Model	MAE	RMSE	R2
XGBoost	0.9458	1.3065	0.4103
LightGBM	0.9508	1.3138	0.4041
CatBoost	0.9488	1.3139	0.4043
CNN-LSTM	**0.5056**	**0.6582**	**0.7905**
Bi-LSTM	0.5357	0.7372	0.2487
Transformer-based model	0.6479	0.8881	0.6931

The CNN-LSTM model again demonstrated the strongest performance, achieving the lowest error metrics and the highest explanatory power (R^2 = 0.7905). This suggests that the hybrid architecture is particularly effective at analysing the spatio-temporal dynamics inherent in multi-source data. By extracting spatial patterns through the CNN layers before processing them with the LSTM, the model maintains a significant advantage in handling the structural complexity of the CDI.

The Transformer-based model showed a notable improvement in this experiment compared to its performance on SPEI, where R^2 was approximately 0.25. With the CDI, it achieved an R^2 of 0.6931. This suggests that the Transformer's attention mechanism is better suited to capturing long-range dependencies within multivariable, aggregated signals than within pure, high-frequency meteorological time series data. Conversely, the Bi-LSTM model performed the weakest in this configuration (R^2 = 0.2487). While it achieved competitive MAE scores, its inability to model the variance within the complex CDI signal underscores a limitation of pure recurrent architectures for multi-index tasks.

To address the inherent class imbalance within the drought score distributions, this study employed weighted averages, a methodology established by Winkler and Clemen (1992) to improve the reliability of predictive aggregation. Oversampling techniques were evaluated as a potential alternative to mitigate data skew, but they yielded only marginal performance gains. This outcome reinforces the observation that while deep learning architectures possess high predictive power, their sensitivity to underlying data distributions requires further investigation (Chandra, 2025).

Comparative analysis of the top-performing CNN-LSTM model shows that the SPEI consistently outperforms the CDI across all statistical benchmarks. Specifically, the SPEI-based model demonstrates the highest level of predictive precision and stability, as indicated by a lower error rate (MAE of 0.27 vs. 0.50 for CDI) and reduced sensitivity to outliers (RMSE of 0.37 vs. 0.65 for CDI). Additionally, the SPEI model achieves a higher degree of correspondence, explaining a greater proportion of variance (R^2 = 0.82) compared to the CDI model (R^2 = 0.79).

These results suggest that integrating temperature-driven moisture deficits within the SPEI provides a more coherent signal for deep learning feature extraction than the multifaceted components of the CDI, particularly in moisture-stressed arid and semi-arid environments (Bozorgi *et al.*, 2025; Winkler and Clemen, 1992). The performance of the SPEI model is attributed to the high signal-to-noise ratio inherent in meteorological data, where precipitation and temperature follow strong, identifiable physical laws (Timmermans *et al*., 2012). In contrast, the CDI model faced structural complexity due to the integration of multi-source satellite and climatic data.

10. Conclusion

This study successfully evaluated three traditional machine learning architectures (XGBoost, CatBoost, and LightGBM) and three deep learning architectures (CNN-LSTM, Bi-LSTM, and Transformer) to identify the best-performing drought forecasting framework for the arid and semi-arid lowlands of Afar and Somali regions in Ethiopia. Our findings demonstrate that deep learning models perform better than traditional gradient-boosting algorithms like XGBoost, LightGBM, and CatBoost in predicting multifaceted drought signals.

The CNN-LSTM hybrid architecture emerged as the most precise forecasting tool across all experiments. By effectively merging convolutional layers for spatial feature extraction with LSTM units for long-term temporal dependencies, this model achieved the highest predictive accuracy and stability for both the SPEI and the CDI indices. Notably, the SPEI-based configuration yielded the overall best performance (R^2 = 0.82, MAE = 0.27), suggesting that the high signal-to-noise ratio in meteorological data provides a more coherent target for deep learning feature extraction than the structural complexity of multi-source indices like the CDI.

Furthermore, the results highlight the importance of model–data alignment. While the CNN-LSTM remained consistently superior, the Transformer-based model showed good performance when transitioned from a single-index (SPEI) to a multivariate (CDI) configuration (R^2 increased from 0.25 to 0.69). This underscores the fact that attention mechanisms are uniquely suited for aggregated, high-dimensionality signals where long-range interactions between vegetation health and thermal stress are critical.

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